

Access Control Across Microservices in a Multi-Tenant Architecture — Design Document

Enterprise-grade Access Control System for a multi-tenant, microservices-based SaaS platform.

🔗 All diagrams below are also available as standalone, rendered images (SVG/PNG) in [docs/diagrams/](#) — regenerate with `make diagrams`.

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1. Overview & Scope

We design and (partially) implement an **Access Control System** that sits at the heart of a multi-tenant SaaS platform composed of independent microservices (User Management, Expense, Payroll, Reporting, Workflow, Notification, Invoice). The system answers one central question consistently across every service:

"Can subject *S*, acting in tenant *T*, perform action *A* on resource *R* (with attributes), under the current environment?"

It must do this with strong tenant isolation, fine-grained rules, full auditability, and at a scale of **thousands of tenants / millions of users / high throughput**.

What is built as runnable code (reference implementation)

Component	Built	Purpose
Auth Service	✓	Authentication, JWT issuance/refresh, user/tenant/role-assignment management
Authorization Service (PDP)	✓	Policy Decision Point: RBAC + ABAC evaluation, role/permission/policy CRUD, audit
Shared PEP library	✓	Policy Enforcement Point middleware imported by each microservice
Expense Service	✓	Sample service demonstrating ABAC (amount thresholds, dept ownership, separation of duties)
Payroll Service	✓	Sample service demonstrating sensitive-data RBAC + tenant isolation
Invoice Service	✓	Third sample — proves a NEW service plugs in with zero core changes (imports the PEP, registers <code>invoice:*</code> permissions, attaches an ABAC issue policy)
Streamlit Admin UI	✓	Manage tenants, users, roles, permissions, policies; live decision simulator; audit viewer
PostgreSQL (+RLS) & Redis	✓	Shared DB with row-level security; decision cache + session store
Reporting / Workflow / Notification	Designed only	They generalize identically to the three sample services; documented, not coded, to fit the timebox

This is a deliberate scope decision: **three sample services prove the entire cross-service authorization story** — Invoice specifically demonstrates the "new service = register permissions + import PEP, no core change" claim *in code*; the remaining services are structurally identical.

2. Assumptions

1. **Tenant = Organization.** One company = one tenant. **A user is a global identity that can belong to many tenants** (the Slack/Google/Atlassian pattern): identity/credentials are global, while *participation* in a tenant is a **Membership** that carries that tenant's roles and org-unit. A user always acts within **one active tenant at a time** (the token is tenant-scoped). See §6.1 and §9.
2. **Identity is internal** for the reference implementation (email + password). Enterprise SSO (OIDC/SAML/SCIM) is designed for but stubbed; the JWT contract is identical regardless of IdP.
3. **Scale target** drives design but is not load-tested in the take-home: $\sim 10^4$ tenants, $\sim 10^6$ users, $\sim 10^3\text{--}10^4$ authz checks/sec at peak.
4. **Postgres** is the system of record. **Redis** is a cache/session store (loss-tolerant).
5. **Trust boundary:** the API gateway terminates external TLS and validates the user JWT; internal service-to-service calls are within a trusted network and additionally use signed service tokens.
6. **Eventual revocation is acceptable within a small bound** (cache TTL, default 15s). Truly instant revocation is available via cache-busting (documented).
7. **2-day build budget** — favors clarity and correctness of the core model over breadth of services.

3. Functional Requirements

Identity & Tenancy

- **FR-1** Create/manage tenants (organizations), each with status (`active` / `suspended`) and tier (`pooled` / `enterprise`).
- **FR-2** Manage **global users** (a user is one identity that may belong to multiple tenants) and **memberships** (a user's participation in a tenant, with an org-unit assignment in that tenant's hierarchy). An admin adds an existing/new user to their tenant as a member.
- **FR-3** Authenticate the **global identity** (email + password), then **select/switch the active tenant** → a tenant-scoped JWT; refresh and revoke sessions.

Access Control

- **FR-4** Define **permissions** as (`service`, `resource`, `action`) triples (e.g. `expense:expense:approve`).
- **FR-5** Define **roles** per tenant; grant permissions to roles; support **role inheritance** (hierarchy).
- **FR-6** Assign roles to a user's **membership** in a tenant (many-to-many) — so the same user can hold different roles in different tenants.
- **FR-7** Define **ABAC policies**: a condition (over subject/resource/environment attributes) + effect (`allow` / `deny`) attached to permissions.

- **FR-8** Evaluate a decision: `check(subject, action, resource) → allow|deny` with a **reason**.
- **FR-9 Dynamic management**: role/permission/policy changes take effect without redeployments and within the cache TTL.
- **FR-10 System roles** (e.g. `tenant_admin`) seeded per tenant; org-unit-scoped role assignments.

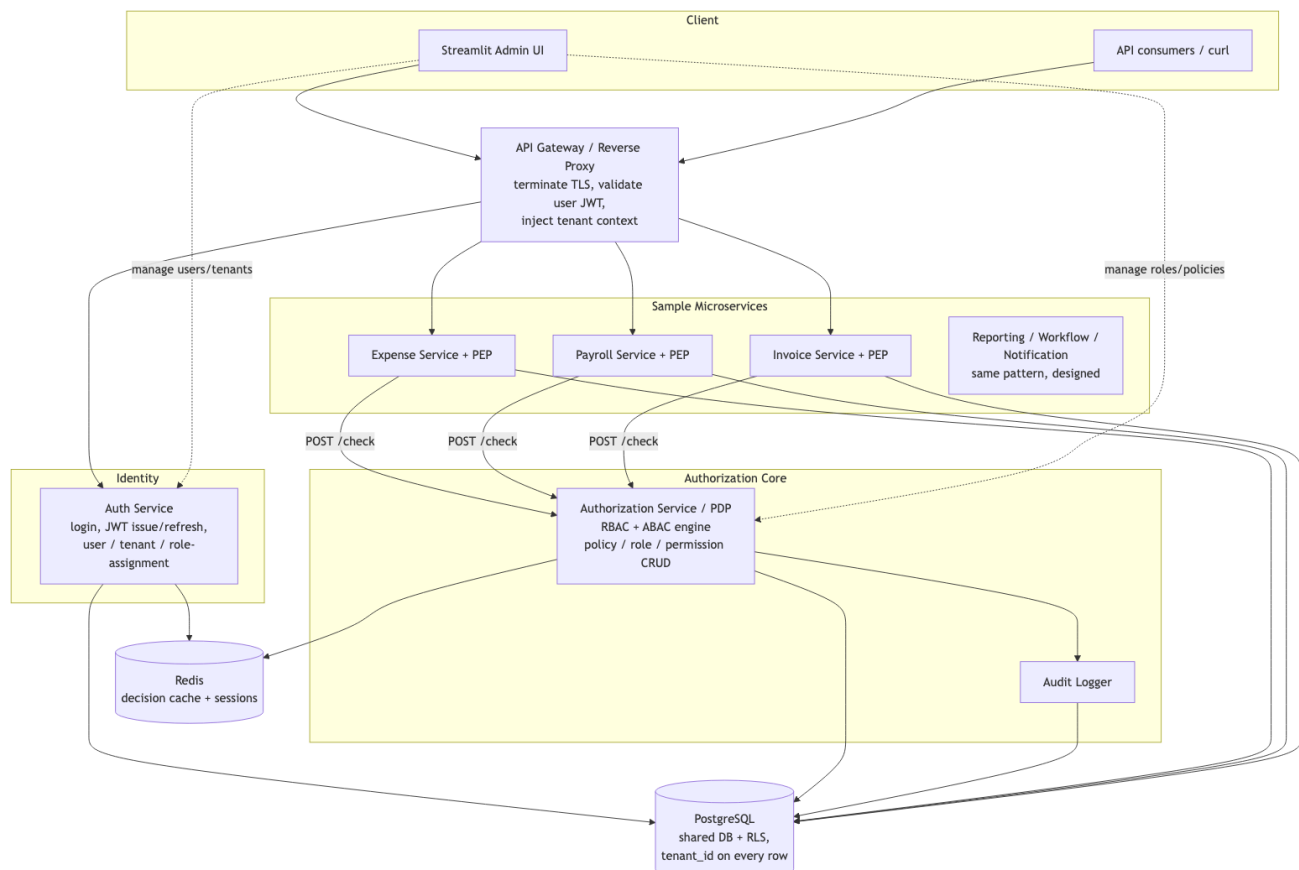
Cross-cutting

- **FR-11** Every privileged action and every authorization decision is **audited** (who, what, when, decision, reason).
- **FR-12** Admin **UI** to perform all of the above and to **simulate** a decision before rolling it out.
- **FR-13** Cross-service authorization: any service can ask the PDP and get a consistent answer.

4. Non-Functional Requirements

#	Requirement	Target / Approach
NFR-1	Tenant isolation	No tenant can read/affect another's data — enforced at JWT, app, and Postgres-RLS layers (defense in depth).
NFR-2	Low-latency authz	p99 <code>check</code> < 10 ms on cache hit; PDP call only on cache miss.
NFR-3	High throughput	Stateless, horizontally scalable services; shared Redis decision cache; read-optimized permission resolution.
NFR-4	Availability	No single hard dependency on the PDP hot path (cache + fail-safe defaults). Target 99.9%.
NFR-5	Auditability/Compliance	Append-only audit log; immutable decision records; supports SOC2/GDPR-style review.
NFR-6	Security	Deny-by-default; explicit-deny-wins; least privilege; signed tokens; hashed secrets.
NFR-7	Consistency of decisions	One decision engine (the PDP) — every service gets identical semantics.
NFR-8	Operability	Structured logs, health checks, metrics, traceable decision IDs.
NFR-9	Extensibility	New service = register permissions + import PEP; no core change.

5. High-Level Architecture



Component responsibilities

- **API Gateway** — single ingress; terminates TLS, validates the user JWT signature/expiry, extracts `tenant_id`, routes to services. (In the reference impl this is a thin layer; in production: Kong/Envoy/APISIX.)
- **Auth Service** — owns identity: tenants, **global users**, **memberships** (user↔tenant), credentials, sessions, role *assignments*. Authenticates the global identity and issues tenant-scoped JWTs (login → select/switch-tenant). Does **not** make authorization decisions.
- **Authorization Service (PDP)** — the brain. Owns roles, permissions, policies. Exposes `POST /check`. Resolves the role graph, evaluates ABAC conditions, writes audit, returns `{decision, reason, policy_id}`. This is also the **PAP** (Policy Administration Point) — CRUD for the access model.
- **PEP (shared library)** — imported by every business service. A FastAPI dependency that (1) validates the JWT, (2) enforces tenant match, (3) consults Redis decision cache, (4) on miss calls the PDP, (5) allows/denies the request. **Enforcement is distributed; decision-making is centralized.**
- **Business services** — own their domain data (expenses, payslips); delegate *all* access decisions to the PEP/PDP.
- **PostgreSQL** — single shared database; `tenant_id` on every business/identity row; **Row-Level Security** as the last line of defense.
- **Redis** — decision cache (key = hash of subject+action+resource+policy-version) and session/refresh-token store.

Why this shape (the central tradeoff resolved)

We separate **authentication** (JWT, slow-changing identity → in the token, zero network calls) from **authorization** (fine-grained, resource-dependent, dynamic → the PDP service, with a decision cache). This gives **token-level speed on the hot path** while preserving **service-level correctness** (instant-ish revocation, ABAC on live resource attributes, full audit). See §15.

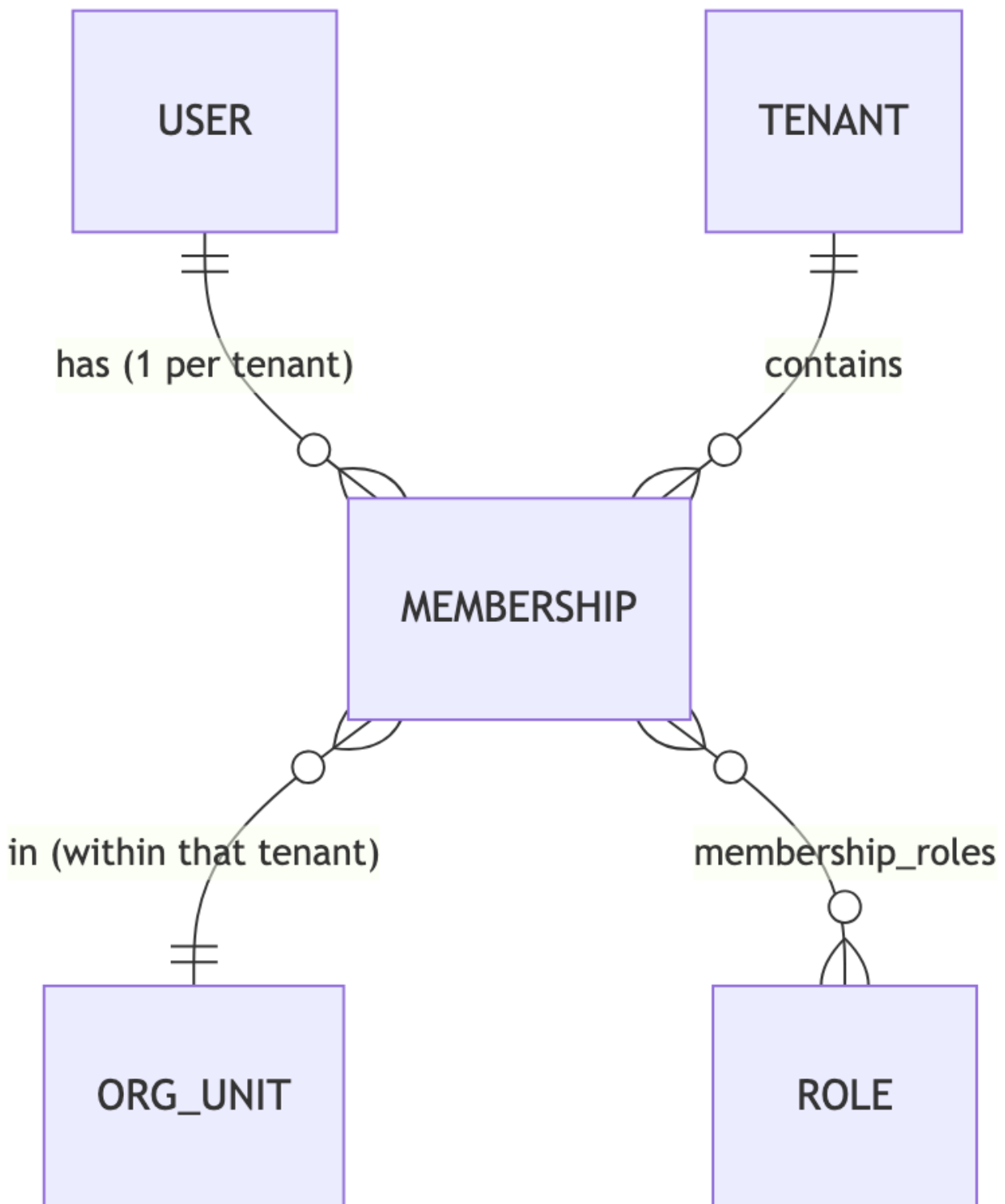
6. Access Control Model (RBAC + ABAC)

We use a **hybrid RBAC + ABAC** model.

- **RBAC** answers "*what is this user generally allowed to do?*" — coarse, role-driven, cacheable, easy to reason about and administer.
- **ABAC** answers "*...but only if these conditions about the specific resource/context hold?*" — fine-grained, dynamic, expressed as policies.

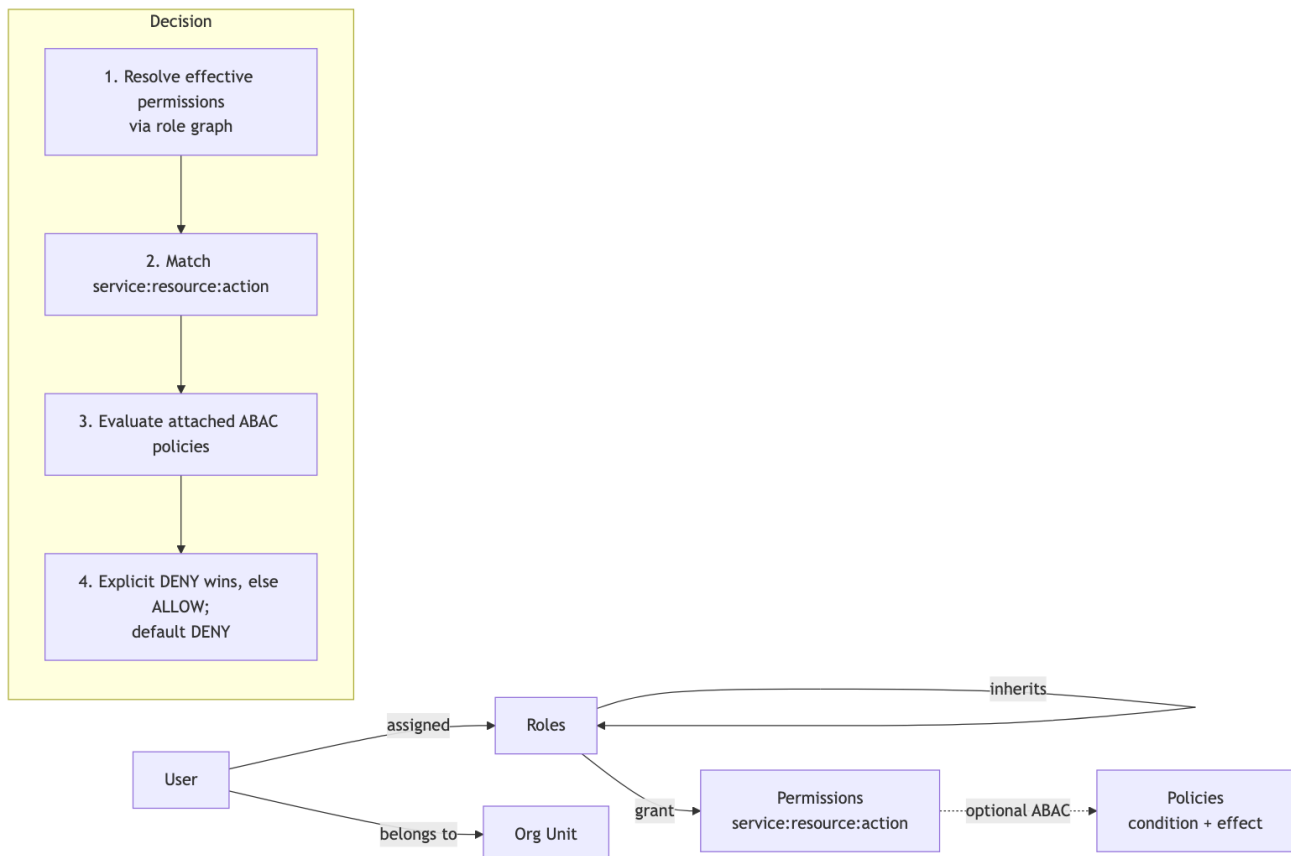
6.1 Multi-tenant membership: identity vs participation

A **user** is a **global principal**; a **membership** is that user's participation in a single tenant. Roles are tenant-scoped, so they attach to the *membership*, not the user — letting one person be `admin` in Tenant A and `viewer` in Tenant B.



- **User (Principal)** — global identity: `email` (globally unique), credentials/MFA, global `status`. Tenant-independent. *Authentication* is global.
- **Membership** — `(user_id, tenant_id, org_unit_id, status)`, unique per `(user, tenant)`; `status` `invited|active|suspended` is **independent per tenant** (suspended in A, active in B).
- **One active tenant at a time** — the access token always carries exactly **one** `tenant_id`. Switching tenants re-issues a token. This keeps the PDP and Postgres-RLS design **completely unchanged** — they always see a single tenant context.
- **Isolation is not weakened** — a token can only ever name a tenant where the user has an *active* membership, re-verified at every issuance/switch; a user can never set the RLS context to a tenant they don't belong to.
- **Adding a member** (in scope): an admin adds an existing-or-new user to their tenant, creating an *active* membership directly. Full invitation/accept emails, self-service domain-join, SSO JIT provisioning, and per-tenant identity linking are deliberately deferred (§17).

- **Cross-tenant platform admin** is a distinct concept (a global principal with platform-level grants), modeled deliberately rather than as a side effect of membership — also future work.



Core concepts

- **Permission** — a (service, resource, action) triple, e.g. `expense:expense:approve`, `payroll:payslip:read`. Global catalog (not tenant-scoped) so semantics are uniform; tenants grant subsets via roles.
- **Role** — a tenant-scoped named bundle of permissions. Roles can **inherit** other roles (`manager inherits employee`), forming a DAG that also models organizational seniority.
- **Org Unit** — node in the tenant's org tree (Company → Dept → Team). A *membership* belongs to one (so the same user can sit in different org units across tenants). Used by ABAC (`resource.dept == subject.dept`) and for scoped role assignments.
- **Policy (ABAC)** — { condition, effect } attached to a permission. The condition is a small JSON boolean expression over `subject.*`, `resource.*`, `environment.*` attributes. Effect is `allow` or `deny`.
- **Direct grant (per-user)** — a permission attached **directly to a membership** (`membership_permissions`), in addition to whatever its roles grant. The effective permission set at decision time is *roles (incl. inheritance) U direct grants*. This is the supported way to give one user one extra permission without inventing a single-member role; it only **widens** the RBAC gate — ABAC deny policies still run afterwards and take priority, so a direct grant can never override an explicit deny. Like roles, direct grants are tenant-scoped (RLS) and changing them bumps the tenant authz epoch (cache invalidation).

Condition language (JSON DSL)

A minimal, safe, serializable expression tree — **no arbitrary code execution** (unlike raw Rego/eval):

```
{ "all": [
  { "lt": ["resource.amount", 10000] },
  { "eq": ["resource.dept_id", "subject.dept_id"] },
  { "neq": ["resource.created_by", "subject.user_id"] }
]}
```

Supported operators: `all` (AND), `any` (OR), `not`, `eq`, `neq`, `lt`, `lte`, `gt`, `gte`, `in`, `contains`. Operands are either literals or dotted attribute paths resolved from the request context. This keeps ABAC **expressive enough** for real enterprise rules while remaining **auditable, cacheable, and injection-safe**.

Decision algorithm (PDP)

```
decision(subject, action_triple, resource_attrs, env):
    1. effective_perms = resolve_role_graph(subject.roles)      # transitive closure
      u direct_grants(subject.user)                          # per-user membership_permissions
    2. if action_triple not in effective_perms: return DENY("no_permission")
    3. policies = enabled_policies_for(action_triple, subject.tenant)
    4. if any DENY policy evaluates true: return DENY(policy)  # explicit deny wins
    5. if ALLOW policies exist for this permission:
      - allow if any of them matches, else DENY("no_allow_policy_matched")
    6. else (no ALLOW policies attached): return ALLOW("rbac") # RBAC baseline grant
```

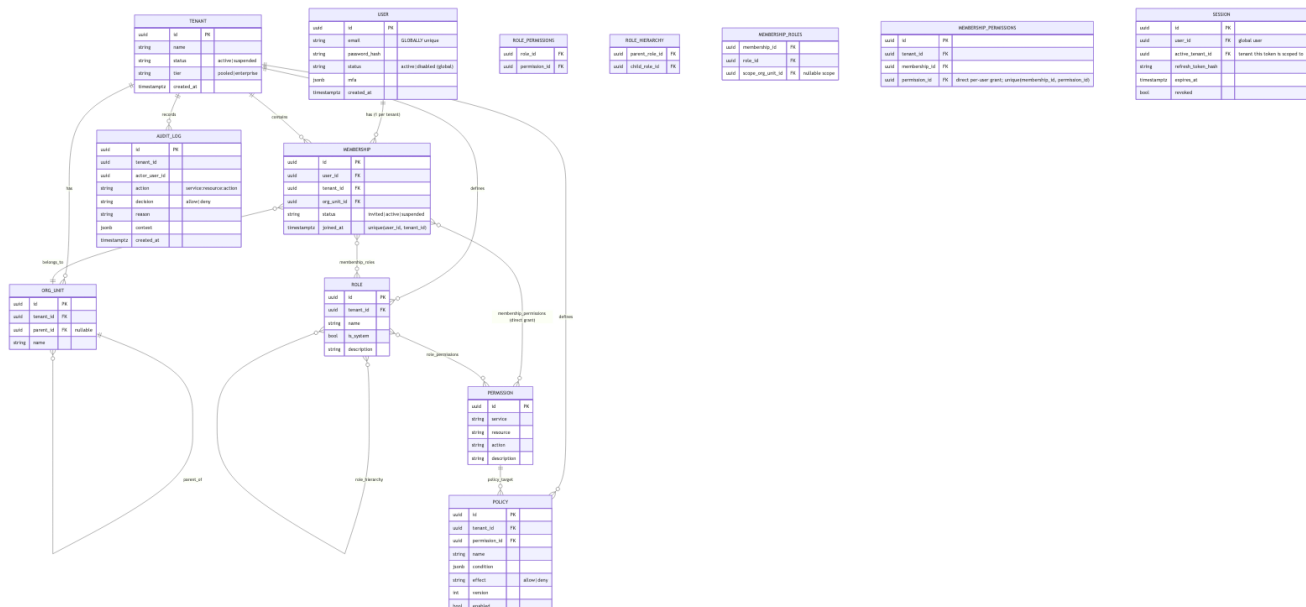
Semantics note: ALLOW-policies act as *additive constraints* on an RBAC-granted permission — if any exist, at least one must match. DENY-policies are *subtractive* and always win. This makes "RBAC grants the capability; ABAC narrows it" precise, and avoids a lone non-matching DENY-policy silently revoking a valid RBAC grant.

Auditing: the PDP records every *decision computation* (i.e. on a cache miss) to `audit_log`. Per-request access logging (including cache hits) is the PEP/gateway's structured log — separating "decision events" from "request volume" keeps the cache valuable without losing the decision trail.

Worked examples (Expense service): - *Employee submits an expense* → has `expense:create` via `employee` role → **ALLOW** (RBAC only). - *Manager approves a \$5k expense in their own dept, not their own* → has `expense:approve`; policy `amount<10000 AND same_dept AND approver≠creator` → **ALLOW**. - *Manager approves a \$50k expense* → permission present but ABAC `amount<10000` fails → **DENY** (reason = policy id). Demonstrates fine-grained control beyond roles. - *Manager tries to approve their own expense* → separation-of-duties `neq(created_by, user_id)` fails → **DENY**. - *Employee with a direct `expense:approve` grant approves a colleague's \$5k same-dept expense* → permission present via `membership_permissions` (not a role); ABAC `expense_approval_limit` matches → **ALLOW**. The same employee approving *their own* expense still → **DENY** (separation-of-duties): the direct grant widens the gate, ABAC still narrows it.

7. Data Model & Schema

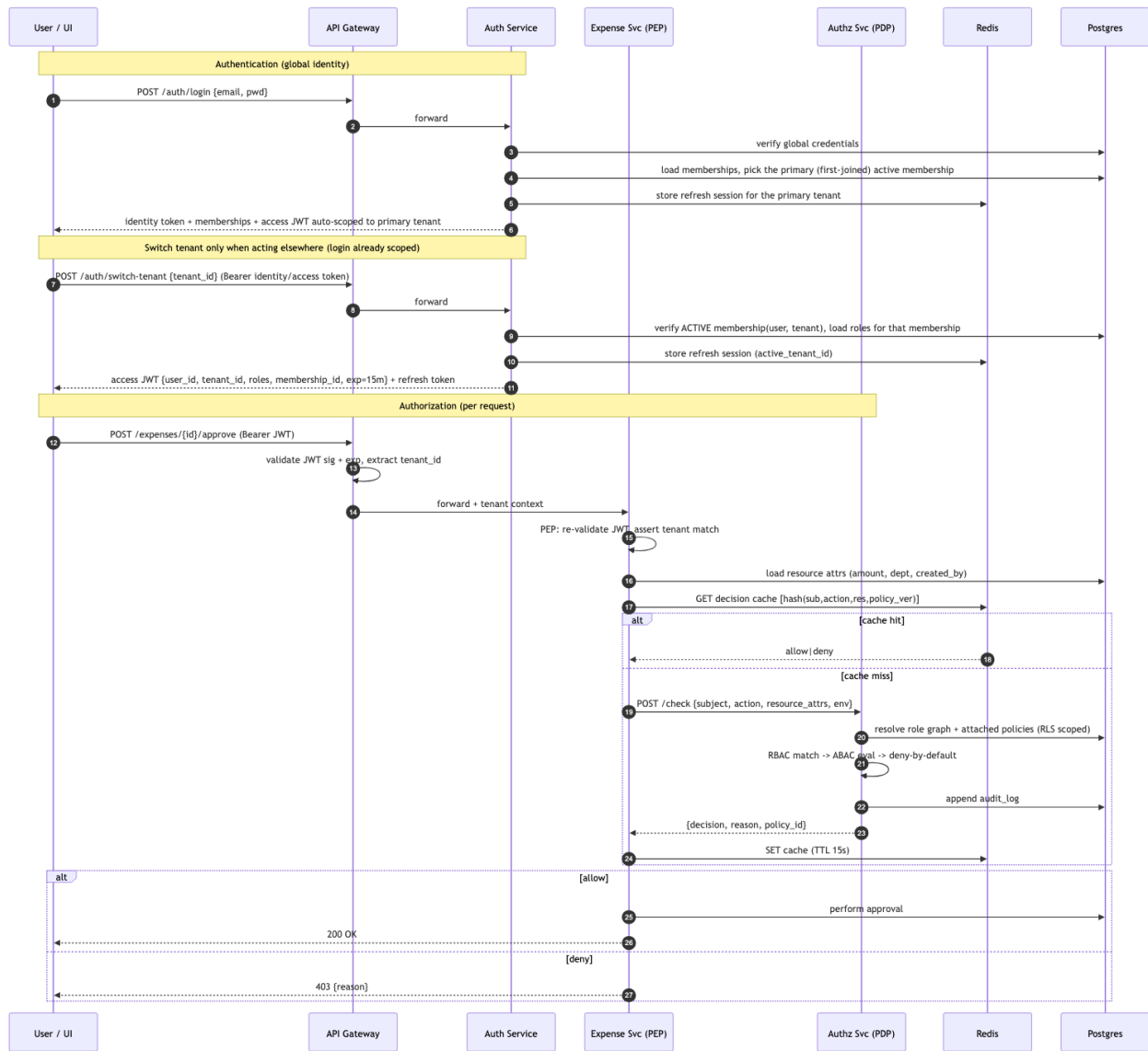
USER is **global** (no `tenant_id`); all tenant-scoped tables carry `tenant_id` and are protected by RLS. The `permissions` catalog is global. A user's link to a tenant is the `MEMBERSHIP` row, which carries that tenant's roles and org-unit.



Notes - USER is **global** (no `tenant_id`); MEMBERSHIP is the only bridge to a tenant. Email is globally unique. A user with no active membership for a tenant can never obtain a token scoped to it. - `permissions` is a **global catalog** (not tenant-scoped) → uniform semantics across tenants; new services insert their permission rows at startup (idempotent). - A per-tenant **authz epoch** (a counter in Redis, bumped on any role/permission/policy change) is included in the **decision cache key**. A single change invalidates *all* cached decisions for that tenant instantly — covering RBAC changes too, not just policy edits. (`policy.version` is additionally kept on each policy row for history/optimistic updates.) - `membership_roles.scope_org_unit_id` enables **scoped grants** (e.g. "manager of Engineering only"); roles are per-membership, so the same user can hold different roles in different tenants. - `membership_permissions` is the **direct per-user**

grant path: a permission attached straight to a membership, unioned with role-derived permissions at decision time. It lets you give one user one extra permission without minting a single-member role; ABAC deny still overrides it. Tenant-scoped (RLS) and epoch-bumped on change like every other authz mutation. - `session.active_tenant_id` records which tenant a refresh session is scoped to; switching tenants issues a new scoped session. - `audit_log` is append-only (no UPDATE/DELETE grants); partitioned by month at scale.

8. Authentication & Authorization Flow



Identity token (post-login, pre-tenant-selection) — minimal; only authorizes `/auth/select-tenant`, `/auth/switch-tenant`, `/auth/me`:

```
{ "sub": "user-uuid", "type": "identity", "iat": 1733300000, "exp": 1733300600 }
```

Access token (tenant-scoped, issued by select/switch-tenant):


```
{
  "sub": "user-uuid",
  "tenant_id": "tenant-uuid",
  "membership_id": "mbr-uuid",
  "roles": ["manager", "employee"],
  "org_unit_id": "ou-uuid",
  "type": "access",
  "iat": 1733300000,
  "exp": 1733300900
}
```

The access token always carries **exactly one** `tenant_id` (the active tenant) — a multi-tenant user gets a fresh token per tenant via `switch-tenant`. Permissions are **not** baked in — only stable identity + roles. Fine-grained permission/policy resolution happens at the PDP so changes are dynamic and revocable.

9. Multi-Tenant Isolation Strategy

Chosen: Shared database, shared schema, `tenant_id` discriminator + PostgreSQL Row-Level Security. Defense in depth across three layers:

Layer 1 – Token
JWT carries tenant_id;
gateway & PEP reject
mismatch



Layer 2 – Application
request-scoped tenant
context; every query auto-
scoped



Layer 3 – Database RLS
SET app.tenant_id per
connection;
policy: USING tenant_id =
current_setting

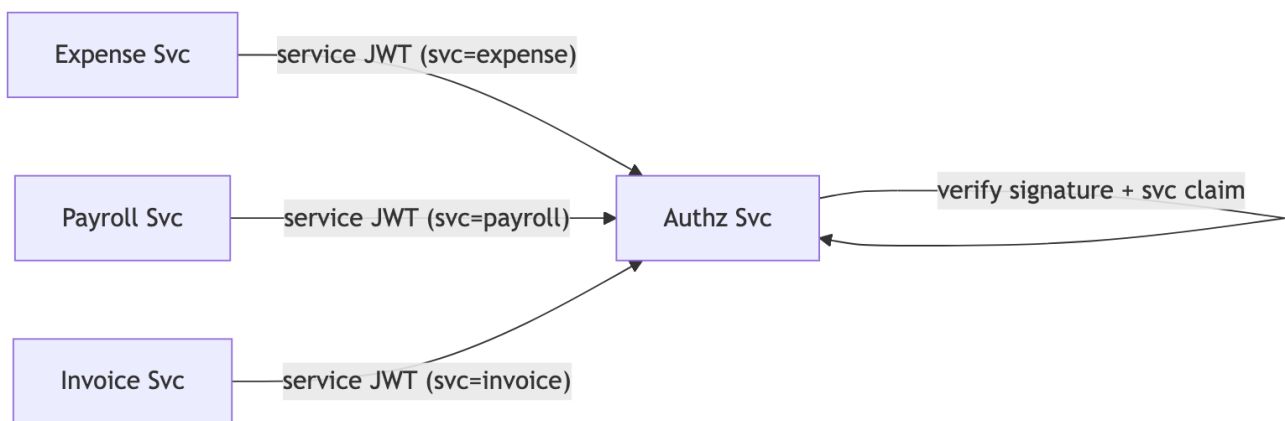
1. **Token layer** — `tenant_id` is a signed claim; the PEP rejects any request whose JWT `tenant` $\neq$ the resource's tenant.
2. **Application layer** — each request opens a DB session and executes `SET app.current_tenant = '<tenant_id>';` ORM models include `tenant_id` and queries are scoped via a shared mixin.
3. **Database layer (the safety net)** — every table has an RLS policy: `sql ALTER TABLE expenses ENABLE ROW LEVEL SECURITY; CREATE POLICY tenant_isolation ON expenses USING (tenant_id = current_setting('app.current_tenant')::uuid);` Even an application bug (forgotten `WHERE tenant_id=`) **physically cannot** leak another tenant's rows, because Postgres filters them.

Why not schema-per-tenant or DB-per-tenant? They give stronger physical isolation but don't scale operationally to thousands of tenants (migrations across thousands of schemas, connection-pool fan-out). RLS gives strong logical isolation at the right cost for this scale. The design supports a **tiered upgrade**: premium/ enterprise -tier tenants can be siloed into a dedicated DB later (`tenant.tier` field already present) without changing application code — the tenant context just points at a different connection.

Cross-tenant access = automatic deny + audit. There is no code path that grants it.

Multi-tenant users do not weaken this. A user belonging to many tenants still acts under a token scoped to exactly one active tenant, and that token is only ever issued after re-verifying an *active* membership. The RLS context (`app.current_tenant`) is derived from the token's `tenant_id` , so a multi-tenant user has no way to read another tenant's data — switching tenants requires a fresh token and a fresh membership check.

10. Service-to-Service Security



- Each service holds a **client credential** and obtains a **short-lived service JWT** (claim `svc` , e.g. `expense`) to call the PDP. The PDP verifies the signature and that the caller is a known service before trusting the forwarded user context.
 - **Audience-scoped tokens** (implemented): a service token carries an `aud` claim (default `authz`) and the PDP **verifies the audience**, so a token minted to call the PDP cannot be replayed against a different service. The calling service (`svc`) is also recorded in every decision's **audit context** (`caller_svc`).
 - **Two distinct identities per request**: the *service* (caller) and the *user* (subject) — the PDP authorizes the user, while authenticating the service.
 - **Production upgrade (documented)**: mTLS between services via a service mesh (Istio/Linkerd) and SPIFFE/SPIRE workload identities; network policies restricting who may call the PDP. The reference impl uses signed service tokens as a lighter equivalent that demonstrates the same trust model.
 - **Token hygiene**: short TTLs, key rotation via a JWKS endpoint, no long-lived shared secrets in business services.
-

11. APIs

Auth Service

Method	Path	Purpose
POST	/auth/login	Authenticate global identity → identity token + memberships, auto-scoped to the primary (first-joined) tenant (returns access + refresh tokens in one call)
POST	/auth/select-tenant	Pick a specific tenant → tenant-scoped access + refresh tokens
POST	/auth/switch-tenant	Re-issue a scoped token for a different tenant the user belongs to
GET	/auth/me	Current identity + memberships
POST	/auth/refresh	Exchange refresh token for new access token
POST	/auth/logout	Revoke session
POST	/tenants	Create tenant
POST	/users	Create a global user (identity only)
POST	/tenants/{id}/members	Add a user to this tenant → <code>active</code> membership (admin "add member")
POST	/memberships/{id}/roles	Assign role to a membership (optionally org-scoped)

Authorization Service (PDP + PAP)

Method	Path	Purpose
POST	/check	Decision endpoint: {subject, action, resource, env} → {decision, reason, policy_id}
GET/POST	/permissions	List / register permissions (global catalog)
GET/POST/PUT/DELETE	/roles	Manage roles + role→permission grants + inheritance
GET/POST/DELETE	/memberships/{id}/permissions	Direct per-user grants (roles U direct grants; ABAC deny still wins)
GET/POST/PUT/DELETE	/policies	Manage ABAC policies
GET	/audit	Query audit log (tenant-scoped)

All PAP (admin) endpoints above require a tenant-scoped access token whose holder is a `tenant_admin` of that tenant (`require_tenant_admin`); `/check` accepts a service (PEP) token or a user token. Every admin change is audited. See §13.1.

Example POST /check :

```
// request
{
  "subject": { "user_id": "u1", "tenant_id": "t1", "roles": ["manager"], "dept_id": "d1" },
  "action": "expense:expense:approve",
  "resource": { "amount": 5000, "dept_id": "d1", "created_by": "u2" },
  "environment": { "ip": "10.0.0.4", "time": "2026-06-04T10:00:00Z" }
}
// response
{ "decision": "allow", "reason": "policy:expense_approval_limit", "decision_id": "dec-abc", "cached": false }
```

Business services (illustrative — Expense)

Method	Path	Guarded by
POST	/expenses	expense:expense:create
POST	/expenses/{id}/approve	expense:expense:approve + ABAC
GET	/expenses	expense:expense:read (auto tenant-scoped)

Full request/response examples live in [docs/api-examples.md](#).

12. Scalability & Reliability

Throughput / latency - All services are **stateless** → scale horizontally behind the gateway. - **Decision cache** (Redis) absorbs repeated checks; key includes `policy.version` so it's both fast and correct. p99 < 10 ms on hit. - Permission resolution is read-heavy and cache-friendly; role graphs are small and can be memoized per request. - Postgres scaled via read replicas for `check/audit` reads; connection pooling (PgBouncer).

Reliability - **Fail-safe defaults**: if the PDP is unreachable on a cache miss, the PEP **denies** (secure default) for sensitive actions; a configurable fail-open is available only for explicitly low-risk read endpoints. - **No SPOF on the hot path**: cache hits don't touch the PDP; PDP runs N replicas. - **Graceful degradation**: Redis down → fall back to direct PDP calls (slower, still correct). - **Audit durability**: decisions are written before the response is returned for privileged actions.

Scaling the data tier - **Bounded responses**: every list endpoint is paginated (`limit / offset` , hard-capped at 200), so one request can never materialize an unbounded result set. - `audit_log` partitioned by month + archived to cold storage. - Tiered isolation: heavy `enterprise` tenants siloed to dedicated DBs to remove noisy-neighbor risk.

13. Security & Compliance

- **Deny by default; explicit deny wins.** No implicit grants anywhere.
- **Least privilege**: users get roles, roles get minimal permissions, ABAC narrows further.
- **Credential security**: passwords hashed (bcrypt/argon2); refresh tokens stored hashed; access tokens short-lived (15m).
- **Token security**: signed JWT (asymmetric in prod via JWKS), audience/issuer checks, tenant-claim binding.
- **Injection-safe ABAC**: JSON DSL evaluated by a safe interpreter — **no eval /Rego sandbox-escape surface**.
- **Audit & compliance**: append-only, immutable audit trail of every decision and admin change → supports SOC 2, ISO 27001, GDPR access reviews. Each decision has a `decision_id` for traceability.
- **Data isolation**: RLS guarantees tenant boundary at the storage layer.
- **Separation of duties** modeled natively via ABAC (`creator ≠ approver`).

13.1 Admin-surface authorization (hardened)

The Auth service is the **identity authority** and runs under the `identity_user` role which has **BYPASSRLS**; the Authz **PAP** (roles/permissions/policies/direct-grants) is tenant-scoped but is the surface that *defines* authorization. Neither is protected by RLS against a *member of the tenant*, so protection must come from an application-level check on the caller.

The gap (now closed): previously every PAP and management endpoint was guarded only by `require_token / require_access` — "*is the caller authenticated / tenant-scoped?*", **not** "*is the caller an admin?*". That was a **vertical privilege-escalation** hole: any tenant member could create roles, write policies, assign roles, or **directly grant themselves a permission** (`POST /memberships/{id}/permissions`) — a one-call self-escalation.

Fixes applied: 1. `require_tenant_admin` (both services) — privileged endpoints now require a tenant-scoped access token whose holder has the `tenant_admin` role. Applied to the **entire Authz PAP** (roles, permissions, policies, direct grants, audit) and the **Auth management** endpoints (members, role assignment, org-units, user/tenant create+list). Endpoints that name a tenant in the path also assert the **path tenant == the token's tenant** (`_assert_tenant`), and role assignment additionally verifies the target membership belongs to the caller's tenant — so an admin of tenant A cannot act on tenant B. 2. `/check subject binding` — the PDP no longer trusts a client-supplied subject on the user path. A **regular member's** subject is bound to their own token (they cannot fabricate `user_id / roles` to probe or shape decisions); a `tenant_admin` may still submit an arbitrary subject *pinned to their own tenant* (the decision **simulator**, which grants nothing they don't already have); the **service (PEP)** path is trusted, since the PEP builds the subject from a verified user token. 3. **Admin-change auditing** — every PAP/management mutation now writes an `audit_logs` row (`decision="info"` , e.g. `admin:role.grant_permission` ,

`admin:membership.grant_permission`, `admin:member.assign_role`), so authorization changes — not just decisions — leave a forensic trail.

Residuals (still future work, §17): a true cross-tenant **platform-admin** principal is not yet modeled, so the genuinely global endpoints (`create/list` of tenants and global users) are gated to `tenant_admin` rather than a platform role — this removes the any-member hole but is coarser than ideal. And because role *assignments* are carried in the access token, an admin demoting a user takes effect at the next token refresh (≤ 15 min); direct-grant and policy changes are instant via the epoch.

14. Operational Concerns

Concern	Approach
Monitoring	Per-service liveness (<code>/healthz</code> , process up) and readiness (<code>/readyz</code> , probes Postgres + Redis and returns 503 when a hard dependency is down — so orchestrators stop routing to a degraded pod). Metrics on check latency, allow/deny ratio, cache hit rate, PDP error rate.
Auditing	Central, queryable audit log (UI viewer); every decision carries <code>decision_id</code> , reason, and full context.
Debugging	The decision simulator in the UI replays a <code>check</code> and shows <i>which</i> role granted and <i>which</i> policy decided — explainable authorization. <code>reason</code> / <code>policy_id</code> returned on every deny.
Tracing	Correlation ID propagated gateway → service → PDP → audit.
Change safety	Policy edits are versioned; the UI simulator lets admins test a policy before enabling it.
Onboarding a new service	Register its permissions, import the PEP, attach policies — zero core changes.
Schema management	Bootstrap <code>create_all</code> + RLS/roles install; no Alembic / migration versioning (idempotent create-from-scratch). Limitation and production path in §15.5.

15. Key Tradeoffs & Alternatives Considered

15.1 Access model — RBAC+ABAC vs ReBAC vs pure RBAC vs OPA

Chosen: RBAC + ABAC hybrid. - *vs Pure RBAC*: RBAC alone can't express "approve only if amount < 10k and same dept" — the doc explicitly requires fine-grained/policy rules. ABAC adds exactly that. - *vs ReBAC (Zanzibar/OpenFGA)*: ReBAC is more powerful for deep relationship graphs and is the natural future evolution, but building or operating it well exceeds a 2-day budget and doesn't echo the doc's own "roles/permissions + policies" vocabulary. **Documented as the scale-up path.** - *vs OPA/Rego*: Externalizing to OPA is flexible and battle-tested, but pushes authorization logic into Rego files rather than a modeled, UI-manageable DB — and the assignment wants *dynamic*, *admin-managed* roles/policies. Our JSON DSL keeps policies as **data** (editable in the UI, versioned, auditable) instead of code.

Deep dive: [ZANZIBAR-VS-XACML.md](#) — the long-form **Zanzibar (ReBAC) vs XACML (PDP/PEP + ABAC)** comparison and why we follow the XACML topology.

15.2 Enforcement — central PDP + cached PEP vs token-baked vs pure-central vs sidecar

Chosen: JWT identity + central PDP + distributed PEP with decision cache. - *vs Permissions-in-JWT*: fast but **stale** — can't revoke until token expiry, can't do resource-dependent ABAC (resource attrs unknown at login), and tokens bloat. Disqualifying for an access-control product. - *vs Pure central PDP (call every time)*: correct but a network hop per request and a hot-path SPOF. We keep its correctness and add a cache to remove the latency. - *vs Fully distributed sidecar (OPA bundles)*: lowest latency, but policy-propagation lag and weaker revocation; heavier to operate. Our cache TTL (15s) gives most of the latency win with simpler consistency.

15.3 Tenant isolation — shared+RLS vs schema-per-tenant vs db-per-tenant

Chosen: shared DB + `tenant_id` + RLS, with a tiered upgrade path. Scales to thousands of tenants cheaply with strong *logical* isolation; physical isolation reserved for premium tenants via the `tier` field. (See §9.)

15.4 Revocation latency vs throughput

Cache TTL is the knob. Default 15s balances near-instant revocation with high cache-hit throughput. `policy.version` in the cache key makes **policy edits** effective immediately regardless of TTL; only role-assignment changes ride the TTL (and the UI offers an explicit "invalidate user" cache-bust for emergencies).

15.5 Schema management — bootstrap `create_all` vs versioned migrations (Alembic)

Chosen: imperative bootstrap, no migration tool. The schema is defined by the SQLAlchemy ORM models and materialized at startup by `services/common/bootstrap.py`: `Base.metadata.create_all()` creates any missing tables, then the same script creates the DB roles, grants, and installs the RLS policies. It is **idempotent create-from-scratch**, run once by the `bootstrap` container (`make bootstrap`).

We deliberately did *not* adopt Alembic (or any migration versioning). The reasoning:

- **Fixed schema, fixed timebox.** This is a 2-day reference implementation with a single, stable schema and no evolution story to manage. Versioned migrations earn their keep when a schema changes over time across environments — which is out of scope here. `create_all` plus the RLS/roles bootstrap is the smallest thing that correctly stands the system up.
- **RLS/roles need a post-DDL step anyway.** Enabling/forcing RLS, creating `app_user` / `identity_user`, and granting privileges are not expressible as plain ORM table DDL. Keeping all of it in one explicit bootstrap script makes the *complete* security posture visible in one place, rather than split between autogenerated migrations and hand-written ops SQL.

The trade-off, stated honestly (this is the limitation, not a free lunch):

- `create_all()` only creates **missing** tables. It will **not** ALTER an existing table — no column add/drop/rename, no type changes, no backfills, and **no down-migrations**. Changing the schema today means recreating the database, which is fine for a demo but **not** for production data.
- There is therefore **no migration history** and no safe, reviewable, replayable path to evolve a live database.

Production path (planned, not built): introduce **Alembic** — generate a baseline revision from the current models, fold the roles/grants/RLS steps into migration scripts (or a first data-migration), and replace the `create_all` call with `alembic upgrade head` in the bootstrap/CI flow. This is tracked as future work (§17); none of the application or PEP/PDP code depends on how the schema is applied, so the swap is localized to bootstrap.

16. Requirements Traceability Matrix

Every capability the assignment names, mapped to where it is satisfied.

Assignment phrase	Where addressed
Multiple services (User/Expense/Payroll/Reporting/Workflow/Notification/Invoice)	§1 (Auth + 3 built samples: Expense/Payroll/Invoice; Reporting/Workflow/Notification designed-identical)
Each tenant: own users	FR-2, §6.1 + §7 MEMBERSHIP, §9
A user across multiple tenants (identity vs participation)	§6.1, §7 USER/MEMBERSHIP split, §8 select/switch-tenant
Organizational hierarchy	§6 Org Unit + role inheritance, §7 ORG_UNIT self-ref
Roles and permissions	§6 RBAC, §7 ROLE/PERMISSION
Policies and access rules	§6 ABAC policies + JSON DSL
Authentication and authorization	§8 full flow
Fine-grained access control	§6 ABAC conditions (amount/dept/SoD)
Tenant isolation	§9 three-layer + RLS
Cross-service authorization	§5 PEP/PDP, §11 /check , consistent everywhere
Service-to-service communication	§10 service JWTs / mTLS path
Dynamic role and permission management	FR-9, §6, §12 cache+version, Streamlit UI
Auditability and security compliance	§13, §14, §7 AUDIT_LOG
Thousands of tenants	§9 shared+RLS, §12
Millions of users	§12 stateless scale, indexing, partitioning
High request throughput	§12 decision cache, stateless replicas
Complex enterprise access patterns	§6 RBAC+ABAC, role DAG, SoD, scoped grants
Functional & non-functional requirements	§3, §4
High-level architecture	§5
Auth & authz flow	§8
Multi-tenant isolation strategy	§9
Access control model	§6
Service-to-service security approach	§10
APIs and data models	§7, §11
Scalability & reliability	§12
Security & compliance	§13
Operational concerns (monitoring/auditing/debugging)	§14
Assumptions documented	§2
Tradeoffs explained / alternatives explored	§15
Diagrams (architecture/sequence/schema)	§5, §6, §8, §9, §10

17. Future Evolution

1. **ReBAC for relationship-heavy authorization** — migrate to OpenFGA/Zanzibar-style tuples when org graphs and resource-sharing grow deep; the permission catalog maps cleanly onto relations.
2. **Enterprise SSO** — OIDC/SAML federation + SCIM provisioning; JWT contract already abstracts the IdP. 2a. **Full membership lifecycle** — email invitation/accept flows, self-service domain-join (verified email domain → auto-membership), SSO just-in-time provisioning, per-tenant identity linking, and a deliberate cross-tenant **platform-admin** principal. The **MEMBERSHIP** model and tenant-scoped tokens already support these; only the onboarding plumbing is deferred for the timebox.
3. **Policy-as-data marketplace** — share/version policy templates across tenants.
4. **mTLS service mesh + SPIFFE** — replace service JWTs with workload identities.
5. **Physical isolation tier** — siloed DBs for top-tier tenants (field already present).
6. **Real-time revocation bus** — push cache invalidations (Redis pub/sub) for true instant revocation regardless of TTL.

7. **Versioned schema migrations (Alembic)** — replace the bootstrap `create_all` with a migration history (baseline from current models; roles/grants/RLS folded into migration steps) so a live database can evolve safely. See §15.5.
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End of design document.